Tellus-Onto: An ontology for soil classification and inference in precision agriculture; [Tellus-Onto: uma ontologia para classificação e inferência de solos na agricultura de precisão]

Helfer, Gilson Augusto; Da Silva, Adilson Ben; Bavaresco, Rodrigo Simon; Barbosa, Jorge Luis Victória

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ABSTRACT ORIGINAL

Soil analysis laboratories demand large volumes of data used in precision agriculture. Among them, parameters that represent soil fertility such as texture and organic matter guide the fertilization process. However, this process can take time, thus limiting its usefulness. Therefore, this article proposes an ontology called Tellus-Onto that extends the state of the art in the classification of Brazilian soils according to the organic and textural composition. A series of axioms and semantic rules provided consultations and inferences about their instantiated basis. In order to test the ontology, we added 98 soil sample results and their classifications were inferred precisely and automatically.